Decent Coaching Classes

IX CBSE

Tuesday Holiday Homework

Ch 2: Linear Polynomials | Ch 4: Algebraic Identities | Ch 5: Circles | Ch 7: Probability

Section A — Chapter 2: Introduction to Linear Polynomials

1. Write the degree of each of the following polynomials:

(a) $5x + 7$ (b) $3x^2 - 2x + 1$ (c) 9 (d) $7y^3 - y + 4$

2. Identify which of the following are linear polynomials. Give a reason for your answer:

(a) $2x - 5$ (b) $x^2 + 3$ (c) $4 - 7y$ (d) $x + y + 1$

3. If $p(x) = 4x - 3$, find $p(0)$, $p(2)$ and $p(-1)$.

4. Find the zero of the linear polynomial $p(x) = 3x + 9$.

5. A shopkeeper charges ₹50 as a fixed delivery fee plus ₹20 per kg of goods delivered. Write a linear expression for the total charge C for m kg, and find the charge for 6 kg.

6. Draw the graphs of $y = 2x$ and $y = 2x + 3$ on the same axes and state how the graphs are related. (Use graph/plain paper for the sketch.)

Section B — Chapter 4: Exploring Algebraic Identities

7. Using the identity $(a + b)^2 = a^2 + 2ab + b^2$, expand $(3x + 5y)^2$.

8. Using the identity $(a - b)^2 = a^2 - 2ab + b^2$, expand $(7p - 4q)^2$.

9. Using the identity $a^2 - b^2 = (a + b)(a - b)$, evaluate 105×95 without direct multiplication.

10. Expand $(x + 4)(x + 9)$ using the identity $(x + a)(x + b) = x^2 + (a + b)x + ab$.

11. Factorise: $9x^2 - 25y^2$

12. Simplify using a suitable identity: $(2a + 3b)^2 - (2a - 3b)^2$

Section C — Chapter 5: I'm Up and Down, and Round and Round (Circles)

13. Define the following terms with a rough figure: (a) chord (b) diameter (c) arc (d) concyclic points.

14. State the theorem that relates the perpendicular drawn from the centre of a circle to a chord.

15. Two chords of a circle are equal in length. What can you say about their distances from the centre? Name the theorem used.

16. A chord of length 16 cm is drawn in a circle of radius 10 cm. Find the distance of the chord from the centre.

17. In a circle, an arc subtends an angle of 70° at the centre. What angle does it subtend at any point on the remaining part of the circle?

18. ABCD is a cyclic quadrilateral in which $\angle A = 100°$. Find $\angle C$, and state the property used.

Section D — Chapter 7: The Mathematics of Maybe (Probability)

19. Rank the following events on a scale from 0 (impossible) to 1 (certain), and label each as impossible, less likely, equally likely, more likely, or certain:

(a) The sun will rise in the east tomorrow. (b) It will snow in Mumbai in May. (c) A tossed fair coin shows heads.
(d) You will meet a friend at school tomorrow.

20. A bag contains 4 red, 5 green and 6 blue marbles. One marble is drawn at random. Find the probability that it is (a) red (b) not blue.

21. Write the sample space for tossing two coins together, and find the probability of getting exactly one head.

22. A die is rolled once. Find the probability of getting (a) a number greater than 4 (b) an even number.

23. In 50 trials of tossing a coin, heads appeared 28 times. Find the experimental probability of getting heads. How does it compare with the theoretical probability?

ANSWER KEY

Section A — Linear Polynomials

1. (a) 1 (b) 2 (c) 0 (d) 3

2. (a) Linear (degree 1) (b) Not linear (degree 2) (c) Linear (degree 1) (d) Linear (degree 1, two variables)

3. $p(0) = -3$, $p(2) = 5$, $p(-1) = -7$

4. $3x + 9 = 0 \Rightarrow x = -3$

5. $C = 50 + 20m$; for $m = 6$, $C = 50 + 120 =$ ₹170

6. Both are straight lines with the same slope 2 (parallel lines); $y = 2x$ passes through the origin, while $y = 2x + 3$ is shifted 3 units upward (y-intercept = 3).

Section B — Algebraic Identities

7. $(3x + 5y)^2 = 9x^2 + 30xy + 25y^2$

8. $(7p - 4q)^2 = 49p^2 - 56pq + 16q^2$

9. $105 \times 95 = (100+5)(100-5) = 100^2 - 5^2 = 10000 - 25 = 9975$

10. $(x + 4)(x + 9) = x^2 + 13x + 36$

11. $9x^2 - 25y^2 = (3x)^2 - (5y)^2 = (3x + 5y)(3x - 5y)$

12. Using $a^2 - b^2$ form with $a = (2a+3b)$, $b = (2a-3b)$: $(2a+3b)^2 - (2a-3b)^2 = 4 \times (2a) \times (3b) = 24ab$ [since $(A+B)^2-(A-B)^2=4AB$ with $A=2a$, $B=3b$]

Section C — Circles

13. Chord: a line segment joining two points on the circle. Diameter: a chord passing through the centre (longest chord). Arc: a connected portion of the circle between two points. Concyclic points: points that lie on the same circle.

14. The perpendicular from the centre of a circle to a chord bisects the chord.

15. Equal chords of a circle are equidistant from the centre (Theorem: Equal chords are equidistant from the centre).

16. Half-chord = 8 cm. Distance = $\sqrt{(\text{radius}^2 - \text{half-chord}^2)} = \sqrt{(10^2 - 8^2)} = \sqrt{(100-64)} = \sqrt{36} = 6$ cm.

17. The angle at the centre is double the angle at the circumference on the same arc, so the angle at the remaining part of the circle = $70° \div 2 = 35°$.

18. Opposite angles of a cyclic quadrilateral are supplementary, so $\angle C = 180° - 100° = 80°$.

Section D — Probability

19. (a) Certain ($P = 1$) (b) Impossible/less likely (P close to 0, Mumbai rarely gets snow) (c) Equally likely ($P = 0.5$) (d) More likely (P closer to 1, depending on routine).

20. Total marbles = 15. (a) P(red) = 4/15 (b) P(not blue) = (15−6)/15 = 9/15 = 3/5

21. Sample space = {HH, HT, TH, TT}; n(S) = 4. Favourable outcomes for exactly one head = {HT, TH}, so $P = 2/4 = 1/2$.

22. Sample space = {1,2,3,4,5,6}. (a) Numbers greater than 4: {5,6} $\Rightarrow P = 2/6 = 1/3$. (b) Even numbers: {2,4,6} $\Rightarrow P = 3/6 = 1/2$.

23. Experimental probability = 28/50 = 0.56. Theoretical probability of heads = 1/2 = 0.5. The experimental value (0.56) is close to, but not exactly equal to, the theoretical value — as expected with a limited number of trials.